

APPENDIX TO THE REQUIREMENT GATHERING DOCUMENT (RGD v2.0)

GAP-09

Appendix-I

High Level Design (HLD)

School ERP

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Classification: Confidential — Client & Project Team Only

1. Introduction

This document is Appendix-I to the Requirement Gathering Document (RGD v2.0) for the School ERP System, addressing Gap-09: High Level Design (HLD). It describes the overall system architecture — modules, layers, interactions, and cross-cutting architectural concerns — synthesized from the eight approved appendices (A through H v1.1) that precede it.

This HLD is deliberately technology-neutral: it does not select specific vendors, frameworks, programming languages, or cloud providers except where RGD v2.0 already made an assumption (e.g., cloud-hosted deployment, RGD v2.0 Section 22 Business Assumptions). Where a technology decision is genuinely required and has not yet been made, this document states "Client/Product Decision Required" rather than inventing one.

This document describes the architecture already implied by Appendix-H v1.1's logical data model and the eleven approved functional modules — it does not redesign or reinterpret that baseline. Low Level Design, database SQL, API specifications, sequence/class diagrams, source code, and UI mockups are explicitly out of scope and are deferred to later SDLC phases.

1.1 Intended Audience

- Solution Architects — validating that module boundaries and data flows established in Appendix-D/E/G/H are architecturally coherent before Physical/Low Level Design begins.
- Business Analysts — confirming that every architectural decision traces back to an approved business requirement, with no invented scope.
- Development Team Leads — using this document as the entry point into the more detailed Appendices B through H v1.1 before LLD work starts.
- Project Sponsor / Management — reviewing overall system shape and outstanding technology decisions (flagged Client/Product Decision Required) ahead of budget/vendor approval.

1.2 Document Conventions

- Every reference to a Functional Requirement uses its Appendix-E ID (FR-xx); every Business Rule uses its Appendix-C v1.1 ID (BR-xxx-xxx); every Non-Functional Requirement uses its Appendix-F ID (NFR-xxx-xxx); every entity uses its Appendix-G/H ID (ENT-xxx-xxx).
- The marker "Client/Product Decision Required" appears wherever a specific technology, vendor, or numeric target has not yet been confirmed by the client — consistent with its use throughout Appendices C through H v1.1.
- Diagrams are presented in simple ASCII/text form per the governing instruction for this document, not as UML or vendor-tool exports.

1.3 Relationship to Low Level Design

This HLD is the architectural bridge between the requirement-and-design appendices (A-H v1.1) and the future Low Level Design (LLD) phase. Where this document says a mechanism "exists" or is "reused," it is pointing to a specific, already-approved source (an FR, BR, NFR, or entity) rather than asserting a new architectural fact. Where a genuine architecture decision remains open, it is listed as a Risk (Section 26) or Assumption (Section 24), not silently resolved here.

1.4 Document Structure

This HLD follows the 27-section structure specified for this document. A brief roadmap:

§	Section	Content
1-3	Introduction, Scope, References	Purpose, audience, boundaries, and source documents
4-7	Architecture, Modules, Interaction, Layers	The core structural view of the system
8-9	User Roles, Authentication & Authorization	Who uses the system and how access is controlled
10-14	Database, Integrations, Storage, Notification, Reporting	Major architectural subsystems
15-16	Logging & Audit, Security	Cross-cutting protective architecture
17	Deployment Overview	Logical infrastructure topology
18-20	Scalability, Availability, Performance	Quality-attribute architecture, sourced from Appendix-F
21-22	Institute Isolation, Environments	Open strategic questions and delivery environments
23	Traceability	Confirms every section's source appendix
24-27	Assumptions, Constraints, Risks, Validation	Closing governance content

2. Scope

2.1 In Scope

- Overall system architecture description at a conceptual/logical level
- Module structure and high-level module interaction
- Layered architecture (presentation, business logic, data access, database)
- User roles and authentication/authorization flow at a conceptual level
- High-level database, integration, file storage, notification, reporting, and logging architecture
- Security, scalability, availability, and performance architecture at a conceptual level
- Logical deployment overview and environment strategy

2.2 Out of Scope

- Low Level Design (LLD) — module-internal design, algorithms, pseudocode
- Database SQL / DDL — already explicitly excluded from Appendix-H v1.1 and remains excluded here
- API specifications — endpoint definitions, request/response contracts
- Sequence diagrams and class diagrams — UML-level design detail
- Source code, deployment scripts, and UI mockups/screen designs

2.3 Scope Boundary Rationale

The boundary between this HLD and Low Level Design follows a simple test: if a decision can be made once, at the whole-system level, and remains stable regardless of which module implements it (e.g., "the system uses role-based access control"), it belongs in this HLD. If a decision is specific to a single module's internal implementation (e.g., "the marks-validation algorithm checks X before Y"), it belongs in LLD. This test was applied consistently across all 27 sections.

3. Reference Documents

Document	Role in This HLD
RGD v2.0	Business objectives, scope, stakeholders, NFR baseline, external integrations, regulatory compliance
Appendix-A — Business Process Diagrams (BPMN)	Actor list and process flows underpinning Module Interaction (Section 6)
Appendix-B — Requirement Traceability Matrix (RTM)	FR-to-module mapping
Appendix-C v1.1 — Business Rule Catalogue	Business rules referenced in Security/Audit architecture
Appendix-D — Requirement Dependency Matrix (RDM)	Module dependency chain underpinning Module Interaction (Section 6)
Appendix-E — Functional Requirement Catalogue	Role Permissions (Field 38) underpinning User Roles (Section 8)
Appendix-F — Non-Functional Requirement Catalogue	Direct source for Sections 15-20 (Security, Scalability, Availability, Performance)
Appendix-G — Data Dictionary	Entity/module structure underpinning Database Architecture (Section 10)
Appendix-H v1.1 — ERD & Database Schema Design	Architectural baseline for Section 10; reused, not redesigned

3.1 Reference Document Summaries

Brief characterization of how deeply this HLD draws on each source document.

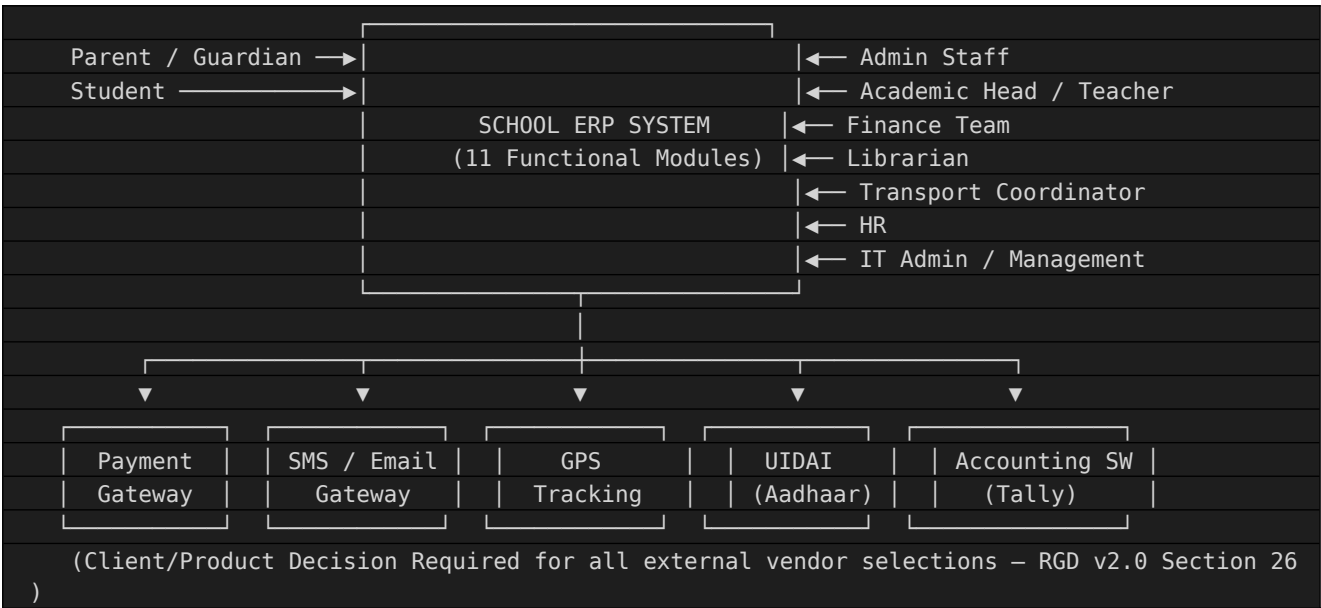
- RGD v2.0 is the ultimate source of business truth for this HLD — every module, objective, and constraint traces back to it, directly or through the intervening appendices.
- Appendix-A (BPMN) supplies the actor lists reused in Section 4.4 and the process-flow logic underpinning the Module Interaction diagram in Section 6, though its detailed swimlane-level flows are not reproduced here.
- Appendix-B (RTM) is referenced implicitly through Appendix-E, since Appendix-E's Field 20 already carries the RTM cross-reference for every FR.
- Appendix-C v1.1 (Business Rules) is the direct source for every Business Rule ID cited in Sections 5, 9, and 16.
- Appendix-D (RDM) is the direct source for the Module Interaction diagram (Section 6) and the Dependency Risk Register items reused in Section 26.
- Appendix-E (FR Catalogue) is the direct source for Role Permissions (Section 8) and the module/FR count table (Section 5).
- Appendix-F (NFR Catalogue) is the single largest direct source for this HLD, underpinning Sections 15 through 20 almost entirely.
- Appendix-G (Data Dictionary) and Appendix-H v1.1 (ERD & Schema Design) together are the direct source for Section 10 and the entity/relationship counts cited throughout.

4. Overall System Architecture

The School ERP is a modular, web-based (and mobile-compatible, per NFR-MOB-001) system centered on a single Student Information Management core, consistent with Appendix-D's finding that the Student entity is the system's structural center. The system serves internal staff actors (Admin Staff, Academic Head, Teachers, Finance Team, Librarian, Transport Coordinator, HR, IT Admin, Management) and external actors (Parents/Guardians, Students) through role-scoped access, with several external systems integrated at the boundary.

4.1 Context Diagram

Shows the system boundary, its actors, and its external integration points, consistent with Appendix-A's actor lists and RGD v2.0 Section 15 (External Integrations).



The system boundary is deliberately drawn to include all 11 functional modules as a single logical system, even though internal actors interact with different modules — no actor communicates directly with an external integration; every external interaction passes through the System boundary first (e.g., a parent's fee payment goes through the Fee Collection module, which then calls the Payment Gateway, rather than the parent's browser calling the gateway directly). This keeps every Business Rule enforcement point (Appendix-C v1.1) inside the trusted system boundary.

4.2 Architectural Style

A layered, modular monolith or modular-services architecture is assumed at this level — the specific choice between a single deployable unit and independently deployable microservices per module is a Physical/Low Level Design decision, not fixed here. Either style is compatible with the module boundaries already established in Appendix-E and the entity boundaries in Appendix-G/H. Extensibility (NFR-EXT-001, Appendix-F) is achieved through the polymorphic association pattern already established in Appendix-H (Section 2.4/11), which allows new modules to attach documents, notifications, and approvals without restructuring existing modules.

5. Major Modules

Reused unchanged from RGD v2.0 Section 12 (Module List) and Appendix-E. Requirement counts are as catalogued in Appendix-E.

#	Module	FR Count (Appx-E)	Primary Owner (Appx-G)
1	Admission & Enrollment	8	Admin Staff
2	Student Information Management	4	Admin Staff
3	Attendance Management	4	Teacher / HR
4	Timetable & Scheduling	3	Academic Head
5	Examination & Result Processing	5	Academic Head
6	Fee Collection	7	Finance Team
7	Library Management	3	Librarian
8	Transport Management	3	Admin Staff / Transport Coordinator
9	HR & Payroll	4	HR
10	Communication & Notifications	3	Admin Staff
11	Reports & Analytics	3	IT Admin / Management

5.1 Module Descriptions

Architectural role, key entities, and primary business rule enforcement for each module, consistent with its Purpose field in Appendix-G and its process flow in Appendix-A.

Across the 11 modules, Appendix-E catalogues 47 Functional Requirements, Appendix-C v1.1 catalogues 71 Business Rules, and Appendix-G/H catalogue 39 core entities (plus 7 out-of-scope stubs carried forward, unexpanded, from Appendix-G). Fee Collection and Admission & Enrollment carry the highest Business Rule density (8 and 8 rules respectively), reflecting their financial and regulatory sensitivity; Library and Transport carry the lowest (6 each), consistent with their comparatively lower risk profile.

5.1.1 Admission & Enrollment

The system's sole entry point for new students. Architecturally owns the Application and SeatAllocation entities (Appendix-G ENT-ADM-001/002) and enforces seat-capacity and RTE-quota business rules (BR-ADM-001 through BR-ADM-008) before a Student record can be created. This module's output — a confirmed Application — is the trigger event for Student Information Management (Section 6.1). It integrates with UIDAI for identity verification (FR-05a) and with UDISE+ for statutory export (FR-05c), both currently Client/Product Decision Required for vendor/API access.

5.1.2 Student Information Management

The structural center of the entire system (Appendix-D Critical Dependency List). Owns the single Student master record (Appendix-G ENT-SIS-001) consumed, directly or indirectly, by every other module. Architecturally, no other module maintains a duplicate copy of student identity data — every cross-module reference is by student_id foreign key, never by a denormalized copy of student attributes.

5.1.3 Attendance Management

Captures daily/period-wise student and staff attendance (Appendix-G ENT-ATT-001/002). Architecturally significant because it gates two other modules: Examination eligibility (via BR-ATT-006) and Payroll processing (via BR-HR-001) — making it one of the highest-fan-out modules in the system despite its conceptually simple data model.

5.1.4 Timetable & Scheduling

Provides the period/subject/teacher schedule (Appendix-G ENT-TT-001) that Attendance and Examination both depend on to know which period is being recorded or examined. Notably, this module has no BPMN diagram in Appendix-A — an inherited Gap-01 dependency carried forward through every subsequent appendix, not a defect introduced in this HLD.

5.1.5 Examination & Result Processing

Owns the marks-entry-to-report-card pipeline (Appendix-G ENT-EXM-001 through 004), enforcing result-integrity rules (BR-EXM-001 through BR-EXM-007) before publication to the parent/student portal. Architecturally the most rule-dense module after Fee Collection, reflecting the academic-integrity sensitivity of its data.

5.1.6 Fee Collection

The highest financial-risk module. Owns invoicing, payment, and refund workflows (Appendix-G ENT-FEE-001 through 005) under strict immutability and authorization rules (BR-FEE-001 through BR-FEE-008). Receives inbound charges from both Library (overdue fines) and Transport (route fees), making it a convergence point in the Module Interaction diagram (Section 6).

5.1.7 Library Management

A lower-sensitivity, largely self-contained module (Appendix-G ENT-LIB-001/002) for catalog and circulation management. Its only cross-module dependency is the fine-posting link into Fee Collection; it does not depend on Attendance, Examination, or any module besides Student Information Management for borrower identity.

5.1.8 Transport Management

Owns route/vehicle allocation and live tracking (Appendix-G ENT-TRN-001 through 003), with a cross-module fee-linkage into Fee Collection (FR-32). Its live-tracking capability (FR-31) depends on an external GPS integration currently Client/Product Decision Required for hardware vendor.

5.1.9 HR & Payroll

Owns the Employee master record and statutory payroll processing (Appendix-G ENT-HR-001 through 005), gated by Attendance closure (BR-HR-001) and enforcing the security-critical access-revocation rule on staff exit (BR-HR-002). This module's Employee entity is the second-most depended-upon record in the system after Student.

5.1.10 Communication & Notifications

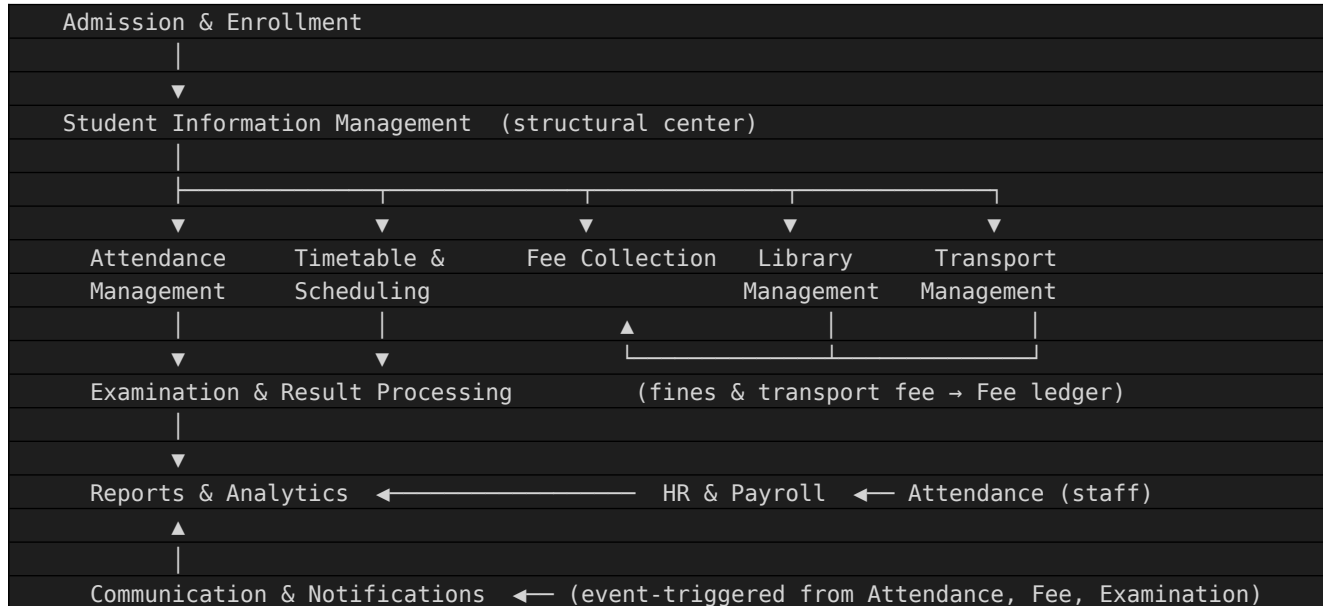
A shared service module (Section 13) consumed by Admission, Attendance, Fee, and Examination for all outbound notifications, rather than an independently-triggered module in its own right. This architectural choice — one shared service rather than four duplicated notification mechanisms — is efficient but creates the cross-module single-point-of-failure risk documented in Section 26.

5.1.11 Reports & Analytics

A pure consumption/aggregation layer (Section 14) with no entities of its own beyond dashboard configuration; the final, terminal stage of the data flow shown in Section 6 — it has no downstream dependents in the entire system.

6. High-Level Module Interaction

Reuses the module dependency chain already established in Appendix-D (Section: Module-Wise Dependency Analysis) and the cross-module relationships confirmed in Appendix-H v1.1 (Section 9).



This diagram is identical in structure to Appendix-D's Module-Wise Dependency Analysis and Appendix-H v1.1's Section 8 module ERDs — reused here as the HLD-level module interaction view, not re-derived.

6.1 Interaction Narrative

- Admission → Student Information Management: a confirmed admission (FR-02) creates the Student record that every downstream module depends on — this is a Mandatory, One-Way dependency (Appendix-D Field 4/11).
- Student Information Management → Attendance, Timetable, Fee, Library, Transport: all five modules require an Active student record before they can operate; none can function on a Draft or non-existent student.
- Timetable → Attendance, Examination: both modules need a published timetable to know which period/subject is being recorded or examined.
- Attendance → Examination: exam eligibility flagging (BR-ATT-006) depends on attendance percentage calculated from Attendance Management.
- Attendance → HR & Payroll: staff attendance closure (FR-12) is a hard precondition for payroll processing (BR-HR-001) — Appendix-D rates this a Very High-strength dependency.
- Library, Transport → Fee Collection: both modules post charges (overdue fines, transport fees) into the same Fee ledger, a converging many-to-one data relationship rather than a peer-to-peer one.
- Examination, Fee, HR → Reports & Analytics: all three modules feed the reporting layer, which itself has no downstream dependents — architecturally, it is always a terminal consumer, never a source.
- Examination, Attendance, Fee → Communication: these three modules trigger notifications as consumers of the shared Communication service (Section 13), rather than Communication independently generating events.

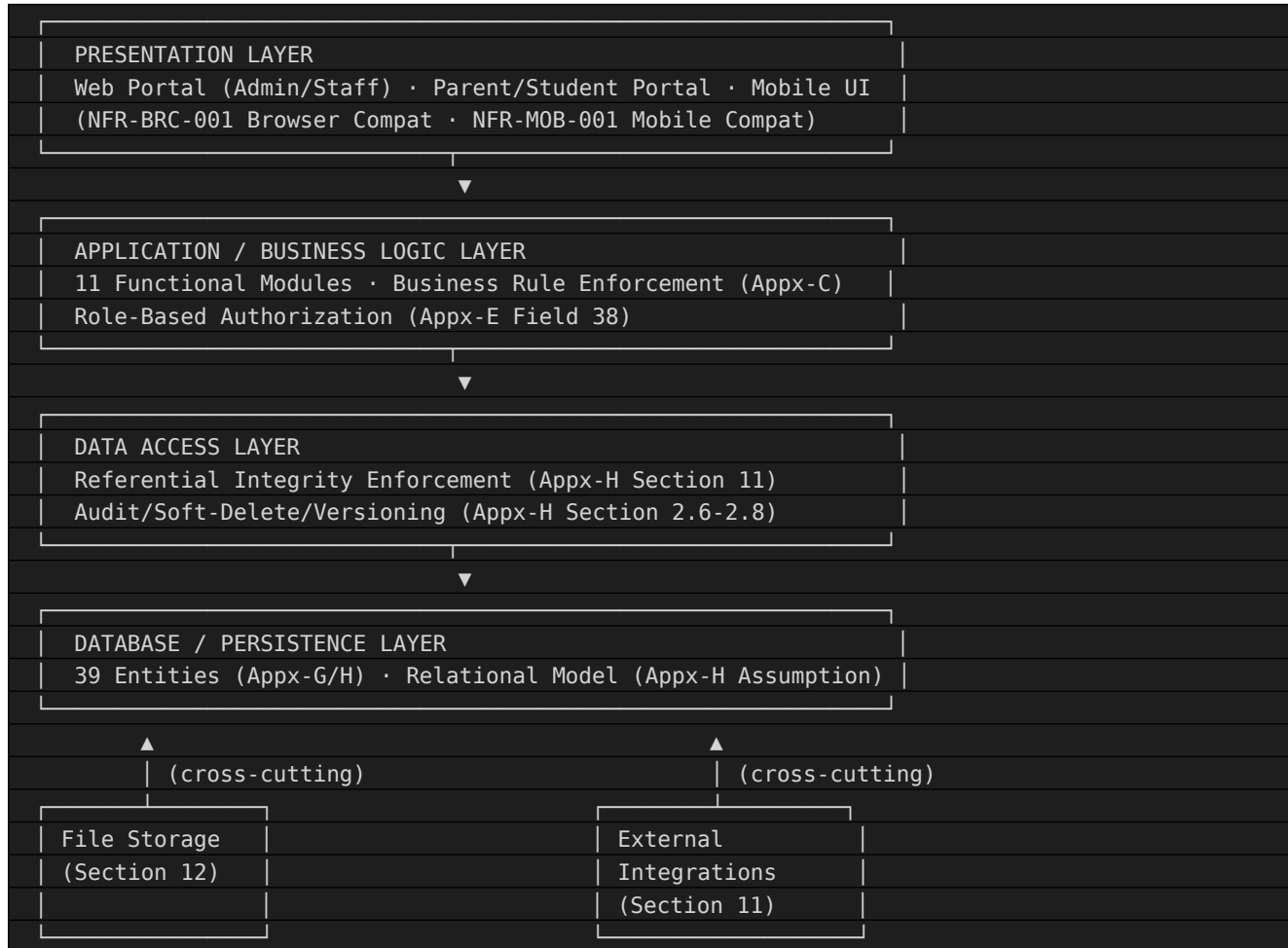
6.2 Implementation Sequence (Reused from Appendix-D)

The logical build/implementation order established in Appendix-D is reused here without modification, since it derives directly from the module dependency chain above.

Phase	Modules	Reason
1 — Foundation	Admission & Enrollment, Student Information Management	Every other module reads from the student record; nothing else can be built first
2 — Master Data & Configuration	Timetable structure, Fee Head configuration, Library catalog, Transport routes, Staff Master	Defines the reference data that Phase 3 transactions are validated against
3 — Core Transactions	Attendance capture, Fee invoicing/payment, Library issue/return, Transport allocation, Payroll processing	Day-to-day operational data creation; depends entirely on Phases 1-2
4 — Academic Processing	Examination scheduling, marks entry, grade/rank calculation, report cards	Depends on both Phase 1 (student/timetable) and Phase 3 (attendance) data
5 — Notifications	Communication & Notifications	Event-triggered shared service; cannot be meaningfully tested until upstream events exist
6 — Reporting	Reports & Analytics	Pure consumption layer; nothing meaningful to report on until earlier phases produce data

7. Layered Architecture

A conventional four-layer architecture is assumed, consistent with the module/entity separation already established across Appendix-E (functional layer), Appendix-F (NFR layer), and Appendix-G/H (data layer). Specific frameworks are not named (technology-neutral).



Cross-cutting concerns — File Storage, External Integrations, Notification dispatch, Logging/Audit — are accessed from the Application layer rather than being layer-specific, consistent with how User, Document, AuditLog, and NotificationLog are modeled as shared entities in Appendix-G/H rather than module-specific ones.

7.1 Layer Responsibilities

Layer	Responsibilities	Governing NFR/BR Source
Presentation	Render role-scoped UI; client-side validation (Field 36, Appendix-E); responsive rendering across desktop/tablet/mobile	NFR-BRC-001, NFR-MOB-001, NFR-ACC-001 through 005
Application / Business Logic	Enforce all 71 Business Rules; orchestrate the Main/Alternate/Exception flows defined per FR in Appendix-E; role-based authorization checks	Appendix-C v1.1 (all rules); NFR-AUTHZ-001
Data Access	Referential integrity, soft-delete, versioning, audit-	Appendix-H v1.1 Sections 2.6-2.8, 11;

Layer	Responsibilities	Governing NFR/BR Source
	column population on every write	NFR-DI-001/002
Database / Persistence	Durable storage of all 39 entities; index-backed query performance; partition-ready for high-volume tables	Appendix-H v1.1 (full); NFR-DBP-001; NFR-SCAL series

7.2 Layer Isolation Rationale

Strict layer separation ensures that a change to the Presentation layer (e.g., adding a new UI theme) cannot silently bypass Business Rule enforcement, and that a change to the Database layer (e.g., adding a partition) cannot silently violate an Application-layer invariant. This is a standard defense against the kind of client-side-only validation gaps flagged as risks in Appendix-E's Validation Rules fields.

7.3 Presentation-to-Application Boundary

The Presentation layer communicates with the Application/Business Logic layer through a well-defined boundary (conceptually an API boundary, though no specific protocol or specification is fixed here per Section 2.2's exclusion of API specifications). This boundary is the single point where Section 9's Authentication & Authorization flow is enforced for every request, regardless of which of the 11 modules or which of the three client types (Admin/Staff web portal, Parent/Student portal, mobile) originates the request.

7.4 Module Boundary Within the Application Layer

Within the Application/Business Logic layer, the 11 modules are organized as internally cohesive, externally decoupled units — a module accesses another module's data only through that module's own entities (per Appendix-H v1.1's Foreign Key relationships), never by reaching directly into another module's internal logic. This is what allows the Physical/Low Level Design phase to choose either a monolithic or services-based implementation (Section 4.2) without this HLD needing to pre-decide that question.

4.3 Key Architectural Principles

- **Single Source of Truth:** Student (Appendix-G ENT-SIS-001) and Employee (ENT-HR-001) are the sole authoritative records for their respective identities; no module maintains a duplicate copy of student or staff data (Appendix-D Critical Dependency List).
- **Shared Services Over Duplication:** Communication (Section 13), File Storage (Section 12), and Audit Logging (Section 15) are single, shared architectural services consumed by all 11 modules, not reimplemented per module.
- **Explicit Business Rule Enforcement:** every Business Rule in Appendix-C v1.1 maps to a specific enforcement point in the Application Layer (Section 7), not left as an unenforced convention.
- **Soft-Delete by Default:** no entity supports hard deletion under normal operation (Appendix-H v1.1 Section 2.7), protecting against accidental data loss across every module uniformly.
- **Configuration Over Code:** policy values already catalogued as pending in Appendix-C v1.1 Section 3.5 (rates, thresholds, SLAs) are architected as runtime configuration (Appendix-G ENT-SYS-005), not hard-coded values requiring redeployment to change.

4.4 Actor Summary

Full role-to-permission detail is in Section 8; this summarizes each actor's primary relationship to the system architecture.

- Parent/Guardian and Student are external, portal-scoped actors with read-mostly access plus limited write actions (messaging, self-service library search) — they never access the Application/Business Logic layer's administrative functions directly.
- Admin Staff, Academic Head, Teacher, Finance Team, Librarian, Transport Coordinator, and HR are internal, module-scoped actors whose access is bounded by their assigned Role (Appendix-E Field 38).
- IT Admin is the only actor with cross-module administrative reach (user/role/configuration management, access revocation), consistent with Appendix-G's Technical Owner field being "IT Admin" for every entity.
- Management/School Management is a read-mostly actor over the Reports & Analytics module (Module 11), with Final Approver authority per RGD v2.0's Approval Matrix rather than day-to-day transactional access.

4.5 Non-Functional Requirement Category Mapping

All 40 NFR categories from Appendix-F are architecturally addressed somewhere in this HLD. This mapping confirms full coverage before the section-by-section detail that follows.

NFR Category (Appendix-F)	Addressed In
Performance, API Performance, Database Performance, Reporting Performance, Search Performance, Print Performance	Section 20 (Performance Considerations)
Scalability	Section 18 (Scalability Overview)
Availability, Reliability, Disaster Recovery, Business Continuity	Section 19 (Availability & Reliability Architecture)
Security, Authentication, Authorization, Session Management, Password Policy, Encryption, File Upload Security, Network Security	Section 16 (Security Architecture)
Audit Logging, Logging	Section 15 (Logging & Audit Architecture)
Data Integrity	Section 10 (Database Architecture, reusing Appendix-H v1.1)
Backup	Section 17 (Deployment Overview)
Usability, Accessibility, Browser Compatibility, Mobile Compatibility	Section 4 (Overall Architecture) and Section 7 (Layered Architecture, Presentation layer)
Caching	Section 17 (Deployment Overview, Cache Tier)
Monitoring, Alerting	Section 19.1 (Monitoring & Alerting Architecture)
Error Handling	Section 7.1 (Layer Responsibilities)
Maintainability, Extensibility, Configurability	Section 4.2/4.3 (Architectural Style, Key Architectural Principles)
Localization, Internationalization	Section 7 (Presentation layer, referenced from RGD v2.0 NFR)
Data Retention, Archiving	Section 10 (Database Architecture) and Section 22 (Environment Overview)
Compliance, Privacy	Section 16 (Security Architecture) and Section 12 (File Storage)

NFR Category (Appendix-F)	Addressed In
	Architecture)

8. User Roles

Reused from Appendix-E Field 38 (Role Permissions) and Appendix-G/H's Role entity (ENT-SYS-002). No new roles are introduced.

Role	Primary Modules	Approval Authority (per Appx-C v1.1)
Admin Staff	Admission, SIS, Transport, Communication	Document verification (BR-ADM-005)
Academic Head	Timetable, Examination, Attendance oversight	Re-evaluation, override approvals (BR-EXM-003, BR-ATT-003)
Teacher	Attendance, Examination (marks entry)	None (Contributor)
Finance Team	Fee Collection	Refund/void authority (BR-FEE-002, exclusive)
Librarian	Library Management	None (Contributor)
Transport Coordinator	Transport Management	None (Contributor)
HR	HR & Payroll	Leave approval, payroll sign-off (BR-HR-004)
IT Admin	Cross-cutting (Users, Configuration, Access)	Access reassignment (BR-HR-002), overrides
Management	Reports & Analytics (read)	Final Approver (RGD v2.0 Approval Matrix)
Parent / Guardian	Portal (read + limited write: messaging)	None (external, portal-scoped)
Student	Portal (read-only, plus Library self-service)	None (external, portal-scoped)

8.1 Role Descriptions

- Admin Staff — the highest-frequency internal user, spanning Admission verification, Student profile completion, Transport allocation, and Circular posting; the default "front office" role.
- Academic Head — the primary academic-approval authority, owning Timetable publication, Exam configuration, and the override/re-evaluation approval chain (BR-EXM-003, BR-ATT-003, BR-SIS-004).
- Teacher — a Contributor role scoped to their assigned classes/subjects only (via TimetableEntry), responsible for attendance marking and marks entry, with no cross-class visibility by default.
- Finance Team — the sole role authorized for refund/void actions (BR-FEE-002, exclusive) and the primary consumer of financial reporting (FR-26).
- Librarian — a self-contained Contributor role for the Library module, with no dependency on or access to other modules' data beyond borrower identity lookup.
- Transport Coordinator — manages route/vehicle configuration; allocation confirmation is shared with Admin Staff per Appendix-G's dual ownership note.
- HR — owns the Employee lifecycle end-to-end, including the security-critical access-revocation trigger on staff exit (BR-HR-002).
- IT Admin — the only role with system-wide administrative reach: user/role management, Configuration entity maintenance, and override/exception approval where no other role is authorized.

- Management — a strategic, read-mostly role over Reports & Analytics, holding final sign-off authority per RGD v2.0's Approval Matrix rather than day-to-day operational access.
- Parent/Guardian — external, portal-scoped; can view their linked student(s)' attendance, results, fees, and communication, and initiate relationship-scoped messaging (BR-COM-001).
- Student — external, portal-scoped, read-mostly, with limited self-service (library search, own issue history per BR relevant to FR-29).

8.2 Role Assignment Model

A User (Appendix-H ENT-SYS-001) holds exactly one Role (Appendix-H ENT-SYS-002) at a time via a Many-to-One relationship (REL-001, Appendix-H v1.1 Section 5). Multi-role scenarios (e.g., a teacher who is also a parent) are handled as two separate User accounts linked to the same underlying person via different owner_type/owner_ref_id values, not as a single account with combined permissions — this avoids privilege-boundary ambiguity.

8.3 Role-to-Module Access Matrix

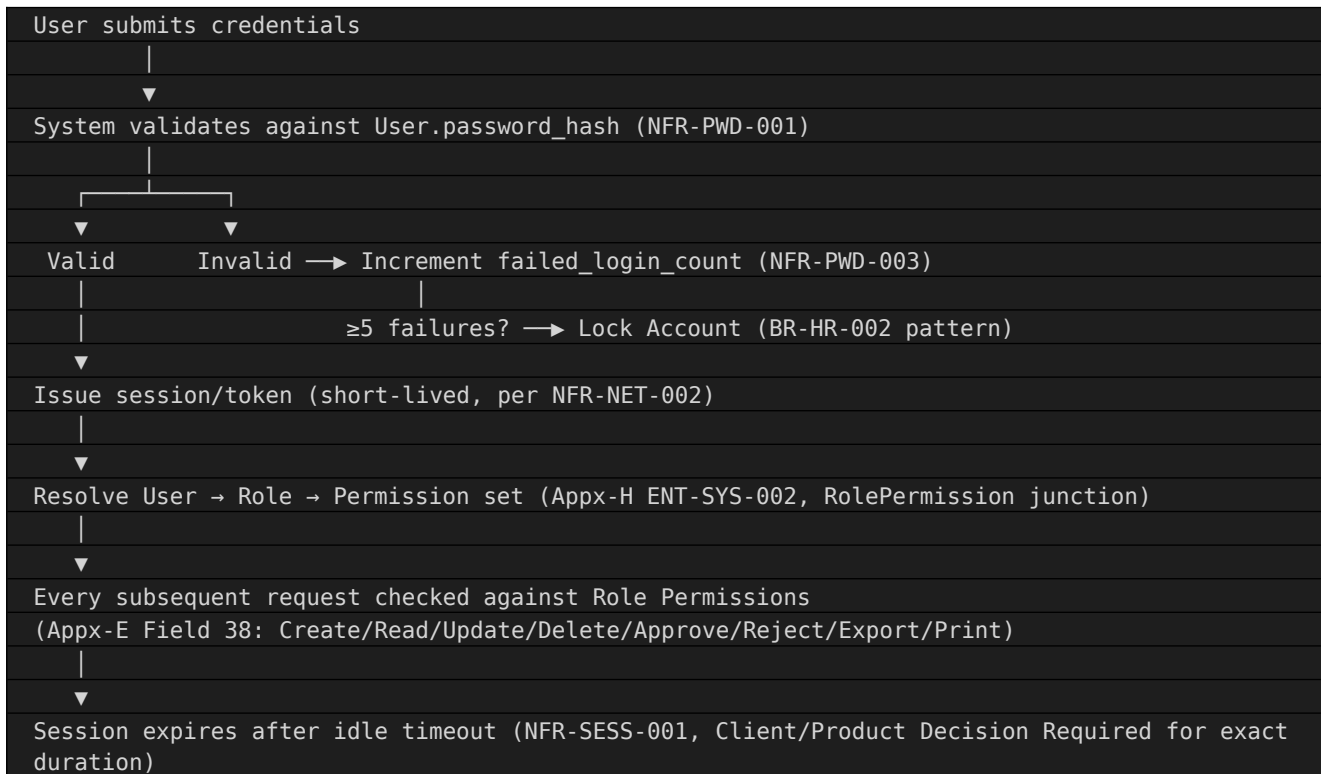
Consolidated view of primary (not exhaustive) access per role per module, reused from the Owner/Approver fields already established in Appendix-G and Appendix-C v1.1. "R" = Read, "W" = Read+Write, "A" = Approval authority, "—" = No routine access.

Module	Admin Staff	Acad. Head	Finance	HR	IT Admin	Parent/Student
Admission	W	R	—	—	R	R (own)
SIS	W	R	—	—	R	R (own)
Attendance	R	A	—	R	R	R (own)
Timetable	R	W	—	—	R	R
Examination	R	A	—	—	R	R (own)
Fee Collection	R	R	W/A	—	R	W (pay)
Library	R	R	—	—	R	R (own)
Transport	W	R	R	—	R	R (own)
HR & Payroll	—	—	R	W/A	R	—
Communication	W	W	R	R	R	W (limited)
Reports	R (limited)	R	R	R	R (full)	—

This matrix is a summary view only; the authoritative, requirement-level permission detail (Create/Read/Update/Delete/Approve/Reject/Export/Print per FR) remains Appendix-E Field 38, which this HLD does not restate in full nor override.

9. Authentication & Authorization Flow

Conceptual flow only, consistent with the User/Role entities in Appendix-H (ENT-SYS-001, ENT-SYS-002) and the Authentication/Authorization NFRs in Appendix-F (NFR-AUTHN-001/002, NFR-AUTHZ-001, NFR-SESS-001, NFR-PWD-001/002/003).



Multi-Factor Authentication is architecturally supported but not enabled at go-live (NFR-AUTHN-002, Appendix-F) — this is a future-enhancement hook point, not a current requirement.

9.1 Password Reset Flow

A conceptual, not implementation-level, flow: user requests reset → system generates a time-limited reset token → token delivered via the Communication service (Section 13) to the user's verified contact → user sets new password subject to complexity rules (NFR-PWD-001) → all active sessions for that user are invalidated, consistent with the session-management posture in NFR-SESS-001.

9.2 Access Revocation Flow (Staff Exit)

Directly enforces BR-HR-002: when an Employee's exit_date is confirmed in HR & Payroll, the linked User account (Appendix-H ENT-SYS-001) is deactivated immediately and automatically, not as a manual follow-up task.

Reassignment of that access (e.g., to a successor) requires a separate, IT Admin-approved grant — access is never silently transferred.

9.3 Authorization Enforcement Points

- API/Application boundary: every request is checked against the requesting User's Role Permissions (Appendix-E Field 38) before any Business Logic executes.

- Field-level: certain fields (e.g., PayrollRun.gross_pay, Student.medical_info) carry additional visibility restrictions beyond basic module access, per the Sensitive/PII classifications in Appendix-G/H.
- Approval gates: elevated actions (refunds, re-evaluations, historical-record overrides) require a resolved ApprovalRequest record (Appendix-H ENT-SYS-007), not merely the requesting user's own role permission.

9.4 Worked Example: Fee Refund Authorization

Illustrates how Sections 8, 9, and 16 combine architecturally for a single sensitive action, reusing BR-FEE-002 from Appendix-C v1.1 exactly as defined there.

- A Finance Team user initiates a refund against a Payment record.
- Authorization check (Section 9.3): the system verifies the requesting User's Role is Finance Team — any other role's request is rejected at this point, per BR-FEE-002's exclusivity rule.
- Business Rule enforcement (Section 7.1): the refund action itself is validated against the Payment's current state (cannot refund an already-refunded transaction).
- Audit logging (Section 15): the refund is written to AuditLog with Who/When/Old Value/New Value, satisfying NFR-AUDIT-001.
- No ApprovalRequest is needed for this specific action, since BR-FEE-002 grants the Finance Team role direct authority rather than routing through a separate approver — illustrating that not every sensitive action requires the Section 9.3 approval-gate pattern, only those explicitly defined as such in Appendix-C v1.1.

10. High-Level Database Architecture

This section summarizes, and does not redesign, the logical database architecture already fully specified in Appendix-H v1.1. Refer to Appendix-H v1.1 for entity-level detail, relationships, keys, and normalization review.

Aspect	Summary (per Appendix-H v1.1)
Model Type	Relational (assumed) — 39 in-scope entities across 11 modules + cross-cutting/system entities
Primary Key Strategy	Single-column surrogate BIGINT keys for every entity
Relationships	45 catalogued relationships; 3 Many-to-Many resolved via junction tables (RolePermission, ClassSubjectMap, StudentGuardianLink)
Normalization	3NF for all entities, with documented JSON-field denormalization for variable-structure data only
Session Scoping	Every session-scoped entity carries an explicit academic_session_id; AcademicSession is the single temporal anchor
Soft Delete	No entity supports hard delete; all deletions are soft (is_deleted flag) with audit trail
High-Volume Entities	AttendanceRecord, MarksRecord, Invoice/Payment, NotificationLog, AuditLog — partition candidates per Appendix-H v1.1 Section 17.8
Institute Isolation	Not Defined in Approved Scope

10.1 Data Distribution Across Modules

Module	Entity Count (Appendix-G/H)	Notable Entities
Cross-Cutting / System	7	User, Role, Guardian, AuditLog, Configuration, Document, ApprovalRequest
Admission & Enrollment	2	Application, SeatAllocation
Academic Master Data	5	AcademicSession, Class, Section, Subject, GradingScheme
Student Information Management	1	Student (structural center)
Attendance Management	2	AttendanceRecord, StaffAttendanceRecord
Timetable & Scheduling	1	TimetableEntry
Examination & Result Processing	4	Exam, MarksRecord, ReportCard, PromotionRecord
Fee Collection	5	FeeHead, FeeStructure, Invoice, Payment, ScholarshipWaiver
Library Management	2	Book, BookIssue
Transport Management	3	Route, Vehicle, TransportAllocation
HR & Payroll	5	Employee, Department, Designation, PayrollRun, LeaveRequest
Communication & Notifications	2	Circular, NotificationLog

10.2 Data Governance Recap

Every entity carries a Business Owner and a Technical Owner (uniformly IT Admin), per Appendix-G Fields 5 and 18 — this HLD does not alter that ownership model. Retention, archival, and purge policy per entity remains as specified in Appendix-G Fields 19-21 and is not restated here.

10.3 Relationship & Junction Summary

Reused directly from Appendix-H v1.1's Final Review (Section 21) and Validation Metrics (Section 20.0).

Metric	Value (Appendix-H v1.1)
Total Entities	39 in-scope + 7 out-of-scope stubs
Total Relationships	45
Total Junction Tables	3 (RolePermission, ClassSubjectMap, StudentGuardianLink)
Total Lookup Tables	20 of 23 requested defined in approved scope
Total Primary Keys	39
Total Foreign Keys	52
Average Relationships per Entity	1.15

10.4 Data Classification Breakdown

Reused from Appendix-G Sections 8-11 (Master/Transaction/Reference/Configuration Data Lists).

Classification	Representative Entities
Master Data	Class, Section, Subject, AcademicSession, GradingScheme, Route, Vehicle, Department, Designation, FeeHead, Book, Role
Transaction Data	Application, AttendanceRecord, MarksRecord, Invoice, Payment, BookIssue, TransportAllocation, PayrollRun, LeaveRequest, Circular, NotificationLog
Reference Data	TimetableEntry (versioned per term)
Configuration Data	Configuration (key-value store, Appendix-G ENT-SYS-005)

11. External Integrations

Reused from RGD v2.0 Section 15. No new integrations are introduced; vendor selection remains pending for all of them.

Integration	Purpose	Status
Payment Gateway	Online fee collection (FR-24)	Client/Product Decision Required — vendor not selected
SMS / WhatsApp Gateway	Notifications (FR-05, FR-11, FR-25, FR-37)	Client/Product Decision Required — vendor not selected
Email (SMTP)	Notifications, report card delivery	Client/Product Decision Required — provider not selected
GPS Tracking	Live bus tracking (FR-31)	Client/Product Decision Required — hardware vendor not selected
UIDAI (Aadhaar)	Identity verification (FR-05a)	Client/Product Decision Required — API access not confirmed
Accounting Software (Tally)	Financial data export	Client/Product Decision Required — integration scope not confirmed
SSO (Google/Microsoft)	Optional staff/student single sign-on	Client/Product Decision Required — desirable, not mandatory

11.1 Integration Pattern

Every external integration is architected as a pluggable, replaceable boundary component rather than a hard-wired dependency, specifically because none of the seven vendors above have been selected yet. Each integration is accessed only from the Application layer (Section 7), never directly from Presentation or Data Access, so a future vendor change requires modifying one integration adapter rather than multiple layers.

11.2 Integration Failure Handling

Each integration has independent health monitoring (NFR-MON-002, Appendix-F), directly addressing the Appendix-D finding that a single shared Communication service (Section 13) could otherwise become a silent, cross-module point of failure. A Payment Gateway outage, for example, is detected and alerted (NFR-ALT-001) independently of an SMS Gateway outage, even though both are consumed by overlapping modules.

11.3 Per-Integration Notes

- Payment Gateway (FR-24): the highest-risk integration architecturally, since it touches financial transaction integrity (BR-FEE-001, BR-FEE-006). Requires idempotency-key support in whichever vendor is selected, to prevent duplicate-charge scenarios during network retries.
- SMS/WhatsApp Gateway and Email (SMTP) (FR-05, FR-11, FR-25, FR-37): together form the Communication module's dispatch mechanism (Section 13); architecturally treated as interchangeable channels behind a single notification abstraction, so a vendor change affects one adapter, not the Communication module's business logic.

- GPS Tracking (FR-31): unlike the other integrations, this one has a hardware dependency (in-vehicle devices) in addition to a software/API dependency, meaning its rollout timeline is coupled to a physical procurement process outside pure software delivery.
- UIDAI/Aadhaar (FR-05a): government-operated integration with its own access-approval process, distinct from a typical commercial vendor selection; timeline risk here is regulatory/administrative, not merely a vendor-selection decision.
- Accounting Software / Tally (FR-26 export): a one-way, outbound data-export integration only — this system is never expected to read financial data back from Tally, keeping the integration boundary simple.
- SSO (Google/Microsoft): explicitly Desirable, not Mandatory, per RGD v2.0 — architecturally, its absence at go-live does not block any other module, since the native User/Role authentication (Section 9) is fully self-sufficient.

12. File Storage Architecture

Reuses the Document entity (Appendix-G/H, ENT-SYS-006), which provides a single storage/reference pattern for all file-based artifacts (admission documents, ID cards, certificates, receipts, report cards) across every module.

- **Storage Model:** object/file storage referenced by a URI (Document.file_path), not stored as database binary blobs.
- **Ownership Pattern:** polymorphic association (owner_type + owner_ref_id) allows any entity — Application, Student, Invoice, ReportCard — to own one or more documents without a separate storage table per module.
- **Access Control:** file access follows the same Role Permission model as its owning entity (e.g., a Payment receipt document inherits Fee Collection's access rules).
- **Storage Location:** cloud-hosted per RGD v2.0 Business Assumptions (Section 22), specific provider Client/Product Decision Required.
- **Security:** sensitive documents (e.g., medical certificates, Aadhaar copies) are subject to the same encryption-at-rest recommendation as their related Sensitive/PII data fields (Appendix-F NFR-ENC-001).

12.1 Document Lifecycle

Follows the same lifecycle pattern as every other entity in Appendix-H v1.1 Section 15: Created (uploaded/generated) → Active → Archived per the owning entity's retention period (Appendix-G Field 19) → soft-deleted only, never hard-purged without a logged IT Admin exception.

12.2 File Type Coverage

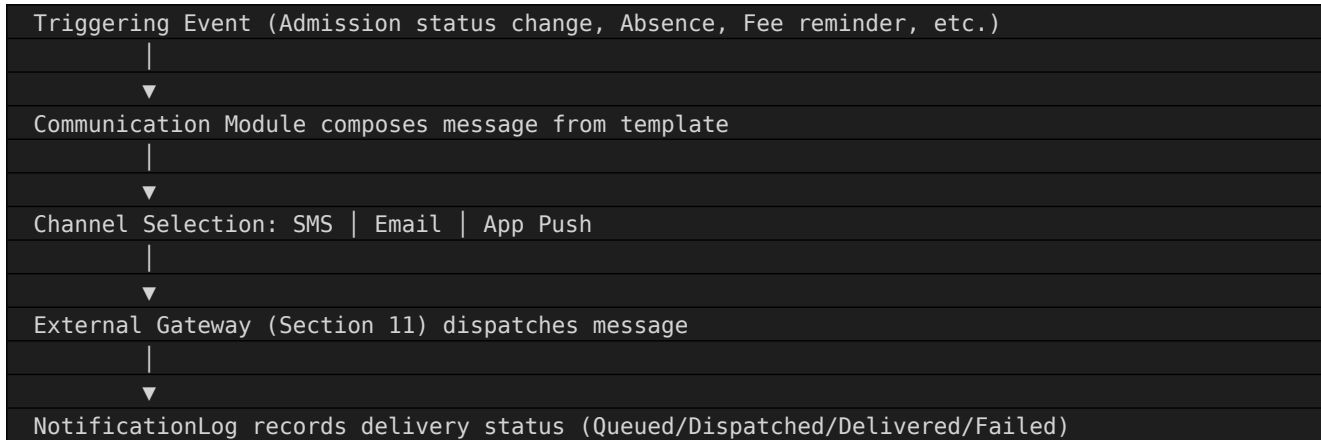
- **Admission documents (FR-01):** birth certificates, prior school records, category proof — uploaded by Parent/Guardian.
- **Generated documents (FR-09, FR-20):** ID cards, transfer certificates, report cards — system-generated from live entity data at generation time, not stale copies.
- **Financial documents (FR-23):** fee receipts — generated per payment, immutable once issued (consistent with BR-FEE-001).

12.3 Storage Access Pattern

Documents are never accessed by direct file-path exposure to the client; every retrieval request passes through the Application layer's authorization check (Section 9.3) first, consistent with the general Presentation-to-Application boundary principle (Section 7.3). This prevents a scenario where a guessed or leaked file path could bypass the Role Permission model.

13. Notification Architecture

Reuses the Communication & Notifications module (Appendix-A Module 9) and the NotificationLog entity (Appendix-G/H, ENT-COM-002), which serves as the single shared notification-dispatch mechanism triggered by multiple other modules (Admission FR-05, Attendance FR-11, Fee FR-25, and direct Communication FR-37).



This shared-service design was flagged in Appendix-D's Dependency Risk Register as a cross-module single point of failure (a gateway outage silently disables notifications from four unrelated modules at once) — mitigated architecturally by independent per-integration health monitoring (Appendix-F NFR-MON-002), not by duplicating the notification mechanism per module.

13.1 Notification Channels & Triggers

Trigger	Source FR	Channel(s)
Admission status update	FR-05	SMS / Email
Student absence	FR-11	SMS / App Push
Fee due / overdue reminder	FR-25	SMS / Email
Result published	FR-21	App Push / Portal
Bulk/targeted circular	FR-37	SMS / Email / App Push
Emergency alert	FR-37 (BR-COM-003)	All channels simultaneously, bypassing standard queue

13.2 Delivery Guarantee Model

Notification dispatch is best-effort with logged failure (NotificationLog.status = Failed), not guaranteed delivery — consistent with Appendix-E Field 32 (System Behaviour) for FR-11, which specifies that a failure is logged and visible, not silently retried indefinitely.

14. Reporting Architecture

Reuses the Reports & Analytics module (Appendix-A Module 10), which is architecturally a pure consumption/aggregation layer with no entities of its own beyond its dashboard configuration — it reads from Fee, Attendance, Examination, and HR data without duplicating it (Appendix-D Implementation Phase 6).

- Role-based dashboards (FR-40) are scoped per Appendix-E Field 38, so a Finance Team user and an Academic Head user see different widget sets from the same underlying platform.
- Custom report builder (FR-41) and trend analytics (FR-42) extend the same base data model rather than introducing a separate reporting datastore.
- Large reports (>10,000 rows, per Appendix-F NFR-RPTP-001) are processed as background jobs rather than blocking the requesting user's session.
- Provisional-data labelling (BR-RPT-004) ensures a report including not-yet-locked data (e.g., unpublished exam results) is visibly marked as such.

14.1 Role-Based Dashboard Scoping

Role	Primary Dashboard Focus
Management	Enrollment, attendance, and revenue summary (FR-40, FR-42)
Academic Head	Attendance and examination performance widgets
Finance Team	Collection summary, outstanding dues, refund tracking (FR-26)
IT Admin	System health and audit-oriented views (Section 15/19)

14.2 Reporting Data Freshness

Every dashboard displays its last-refresh timestamp (NFR-RPT-005/Appendix-F pattern), so stakeholders can judge currency rather than assuming all figures are real-time. This directly supports the Data Refresh Currency Disclosure control already defined in Appendix-F.

15. Logging & Audit Architecture

Reuses the AuditLog entity (Appendix-G/H, ENT-SYS-004) as the single, centralized audit mechanism for the entire system, consistent with Appendix-F's Logging NFR series (NFR-LOG-001 through 010) and NFR-AUDIT-001.

- Every Create/Update/Delete/Approve/Override action across all 11 modules writes to the same AuditLog table (Who/When/Old Value/New Value/IP Address/Reason).
- Module-specific logging (Authentication, Fee Transactions, Attendance, Exam, Library, Transport, HR, Payroll, Configuration Changes) is not a separate mechanism per module — it is the same AuditLog entity, filtered by entity_name (Appendix-F NFR-LOG-010: Consolidated Audit Trail Access).
- Audit records are immutable (write-once) and are never purged without a logged, IT Admin-approved exception (Appendix-G Field 21, Appendix-H Section 15).

15.1 Module-Specific Log Coverage

All logged through the same AuditLog mechanism, filtered by entity_name — not separate logging subsystems per module.

Module	Logged Actions	Governing NFR
Authentication	Login success/failure, lockout events	NFR-LOG-001
Fee Transactions	Payment, refund, credit-note actions	NFR-LOG-002
Attendance	Marks and corrections	NFR-LOG-003
Examination	Marks entry, locks, re-evaluation overrides	NFR-LOG-004
Library	Issue/return/fine transactions	NFR-LOG-005
Transport	Allocation changes, trip logs	NFR-LOG-006
HR	Staff master changes, access revocation	NFR-LOG-007
Payroll	Payroll runs, approvals, adjustments	NFR-LOG-008
Configuration	Fee structure, grading scheme, system config changes	NFR-LOG-009

16. Security Architecture

Directly reuses the 24 Security-related NFRs across 7 categories already fully specified in Appendix-F (Security, Authentication, Authorization, Session Management, Password Policy, Encryption, File Upload Security, Network Security). Summarized here at architecture level only.

Layer	Controls (per Appendix-F)
Network	HTTPS/TLS 1.2+ enforced network-wide (NFR-NET-001); signed, short-lived API tokens (NFR-NET-002)
Application	CSRF, XSS, SQL Injection protections (NFR-SEC-001/002/003); standardized error handling (NFR-ERR-001)
Authentication	Account lockout, MFA-ready architecture (NFR-AUTHN-001/002); password complexity/expiry (NFR-PWD-001/002/003)
Authorization	Role-Based Access Control enforced per Appendix-E Field 38 (NFR-AUTHZ-001)
Data	Encryption at rest for Sensitive/PII fields (NFR-ENC-001); encryption in transit (NFR-ENC-002)
File Handling	Type/size validation and malware scanning on all uploads (NFR-FUS-001)
Audit	Complete, centralized audit trail (NFR-AUDIT-001) — see Section 15

Field-level encryption scope (which specific columns beyond those flagged Sensitive/PII in Appendix-G) remains a Physical Database Design decision, per Appendix-H v1.1 Section 19 Risks.

16.1 Full Security NFR Reference

All 24 Security-family NFRs from Appendix-F, listed here for architectural completeness. Target values are as defined in Appendix-F and not re-derived.

NFR ID	Control
NFR-SEC-001/002/003	CSRF, XSS, SQL Injection protection (OWASP Top 10 baseline)
NFR-AUTHN-001/002	Login mechanism security; MFA-ready architecture
NFR-AUTHZ-001	Role-Based Access Control enforcement (100% of Appendix-E Field 38 boundaries)
NFR-SESS-001	Session timeout (idle expiry, Client/Product Decision Required for duration)
NFR-PWD-001/002/003	Password complexity, expiry, and account lockout
NFR-ENC-001/002	Encryption at rest and in transit
NFR-FUS-001	File upload type/size/malware validation
NFR-NET-001/002	TLS/HTTPS enforcement; API/token security
NFR-AUDIT-001	Complete audit trail (Section 15)

16.2 Security Zones

- Public Zone: unauthenticated admission-inquiry pages and public-facing marketing content (if any) — minimal data exposure, no PII collection without consent capture (NFR-PRIV-001).

- **Authenticated Zone:** the parent/student portal and all staff-facing modules — every request authenticated and authorized per Section 9.
- **Restricted Zone:** Fee Collection financial operations, HR & Payroll compensation data, and Configuration management — subject to the tightest role scoping (Finance Team, HR, IT Admin only respectively).

16.3 Business Rule Enforcement Summary

Reused directly from Appendix-C v1.1's Section 2 (Business Rule Coverage Summary), confirming the scale of business-logic enforcement this architecture must support at the Application layer (Section 7).

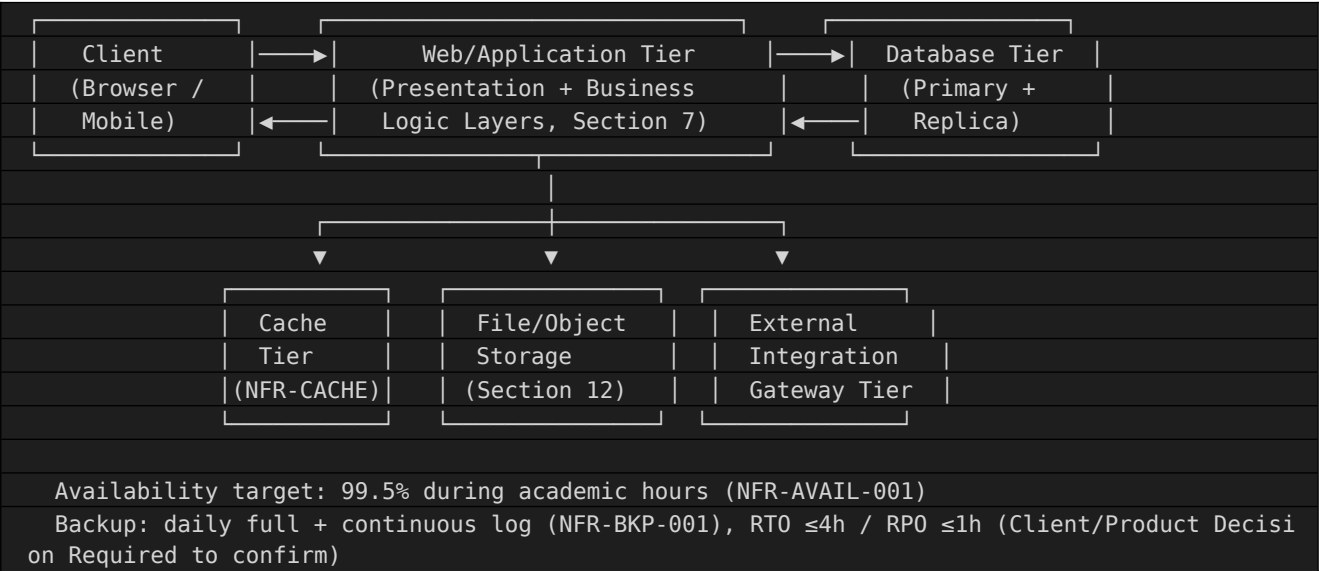
Metric	Value (Appendix-C v1.1)
Total Business Rules	71, across 11 modules
Critical-Priority Rules	19 — require the strictest enforcement and audit posture
Automatic Enforcement Type	57 rules — enforced entirely at the Application/Data Access layer without manual intervention
Manual / Mixed Enforcement Type	14 rules — require a human approval step (Section 9.3 Approval Gates)
Regulatory-Mapped Rules	13 rules — tied to RTE Act, DPDP Act, GST, or statutory payroll law

16.4 Compliance Architecture

Directly supports RGD v2.0 Section 16 (Regulatory Compliance) and Appendix-F NFR-COMP-001. The system's architecture treats regulatory compliance as an enforced, auditable property (via Business Rules and AuditLog) rather than a procedural policy maintained outside the system — e.g., the RTE Act 25% quota ceiling (BR-ADM-003) is enforced at the point of admission confirmation, not verified after the fact.

17. Deployment Overview (Logical Only)

Logical deployment topology only — no specific cloud provider, container orchestration platform, or hosting vendor is named, consistent with the technology-neutral scope of this document. RGD v2.0 assumes cloud hosting (Section 22 Business Assumptions); the specific provider is Client/Product Decision Required.



17.1 Tier Descriptions

- Client Tier: web browser (desktop/tablet) and mobile browser access, per NFR-BRC-001 and NFR-MOB-001 — no native mobile application is assumed unless separately confirmed (Client/Product Decision Required).
- Web/Application Tier: hosts the Presentation and Application/Business Logic layers (Section 7); horizontally scalable to meet the Concurrent Users target (Section 18).
- Database Tier: primary read/write instance plus a replica for read-scaling and availability, consistent with the 99.5% uptime target (NFR-AVAIL-001); partitioning applied per Appendix-H v1.1 Section 17.8 for the highest-volume tables.
- Cache Tier: supports Dashboard/Master Data caching (NFR-CACHE-001) to reduce database load on frequently-read, slow-changing data (Class, Section, Subject, published Timetable).
- File/Object Storage Tier: hosts all Document (Section 12) artifacts, decoupled from the database tier.
- External Integration Gateway Tier: a dedicated boundary layer for all seven external integrations (Section 11), isolating their failure modes from the core application tiers.

17.2 Logical Deployment Notes

This diagram is logical only — it does not specify container orchestration, specific cloud services, or auto-scaling group configuration, all of which are Physical Design/DevOps-phase decisions. The tier boundaries shown are chosen to align directly with the Layered Architecture (Section 7) and the Scalability partition candidates (Section 18), so that a Physical Design team can map logical tiers to concrete infrastructure without re-deriving the boundaries.

17.3 Per-Tier Scaling Considerations

Tier	Scaling Approach	Driver
Web/Application	Horizontal (add instances behind a load balancer)	Concurrent Users target (NFR-SCAL-001)
Database (Primary)	Vertical, with read-replica offload for reporting queries	Query volume from Attendance/Fee/Examination (Section 18)
Database (Partitioned Tables)	Partition-level scaling per Appendix-H v1.1 Section 17.8	AttendanceRecord, AuditLog, NotificationLog, Invoice/Payment volume
Cache	Horizontal, with invalidation-aware distribution	Dashboard/Master Data read frequency (NFR-CACHE-001)
File/Object Storage	Elastic by nature of object storage services	Document volume growth, proportional to Student/Staff/Transaction counts

18. Scalability Overview

Reuses Appendix-F's Scalability NFR series (NFR-SCAL-001 through 008) and Appendix-H v1.1's Partition Recommendations (Section 17.8). Not re-derived here.

Dimension	Target (per Appendix-F, Client/Product Decision Required where noted)
Concurrent Users	2,000 (Client/Product Decision Required to confirm against actual enrollment)
Max Students	20,000 active records (Client/Product Decision Required)
Max Staff	2,000 active records (Client/Product Decision Required)
Attendance Volume	10 million+ records/academic year
Exam Records Volume	5 million+ records/academic year
Fee Transaction Volume	500,000+ transactions/academic year
Partition Candidates	AttendanceRecord, AuditLog, NotificationLog, Invoice/Payment (Appendix-H v1.1 Section 17.8)

18.1 Scalability Design Response

- Horizontal scaling of the Web/Application Tier (Section 17) directly addresses the Concurrent Users target.
- Database partitioning (Section 17.8, Appendix-H v1.1) directly addresses the Attendance, Exam, and Fee Transaction volume targets.
- The single-institute assumption (Section 21) currently bounds total data volume to one institute's scale; multi-institute scaling would require the institute_id extension noted in Section 21.
- Caching (NFR-CACHE-001, Section 17) reduces read load on Master Data (Class, Section, Subject) independent of transactional volume growth, so master-data queries do not degrade even as AttendanceRecord/MarksRecord tables grow to their multi-million-record targets.

18.2 Capacity Planning Horizon

Consistent with Appendix-H v1.1 Section 17.6, capacity planning targets a 5-year data growth horizon without requiring architectural redesign (only infrastructure scaling within the already-defined tier model, Section 17). This means the partition strategy (Section 17.8) and archival policy (Appendix-H v1.1 Section 15) must both be operational from go-live, not added retroactively once volume becomes a visible problem — retrofitting partitioning onto a live, multi-million-row table is materially more disruptive than designing it in from the start.

18.3 Scalability Validation Approach

Load testing against the targets in the table above (Section 18) is the primary validation method for every Scalability NFR, consistent with the Verification Method already specified for each in Appendix-F (e.g., NFR-SCAL-001's verification method is "Load/stress testing"). This HLD does not prescribe a specific load-testing tool, consistent with its technology-neutral scope.

19. Availability & Reliability Architecture

Reuses Appendix-F's Availability, Reliability, Disaster Recovery, and Business Continuity NFR series (NFR-AVAIL-001/002, NFR-REL-001/002, NFR-DR-001/002, NFR-BC-001).

- Target uptime: 99.5% during academic operating hours, measured on a rolling 90-day window (NFR-AVAIL-001).
- Planned maintenance limited to off-peak hours, ≤4 hours/month, with 72-hour advance notice (NFR-AVAIL-002).
- Critical transaction error rate target: <0.1% (fee payment, attendance save, exam lock) (NFR-REL-001).
- Disaster Recovery: documented, drill-tested failover process, at least annual drill cadence (NFR-DR-001/002).
- Business Continuity: manual-workaround procedures documented for Admission, Attendance, and Fee Collection (NFR-BC-001), reusing the Workaround field already defined per requirement in Appendix-D.

19.1 Monitoring & Alerting Architecture

Reuses Appendix-F's Monitoring and Alerting NFR series (NFR-MON-001/002, NFR-ALT-001).

- System Health Monitoring provides real-time visibility into uptime, response time, and error rate for IT Admin (NFR-MON-001), feeding directly into the Availability and Reliability targets above.
- Third-Party Integration Monitoring is independent per integration (NFR-MON-002), directly mitigating the Appendix-D risk that a single shared Communication service could mask which specific external gateway has failed.
- Automated Alerting ensures a critical failure (payment gateway down, backup failure, unusual login pattern) reaches IT Admin within 5 minutes (NFR-ALT-001), rather than being discovered only when a user reports a problem.

19.2 Reliability Design Response

The Reliability targets above (NFR-REL-001/002) are supported architecturally by: transactional consistency enforcement at the Data Access layer (Section 7.1); the atomic-transaction requirement already flagged for the BR-ADM-001/BR-ADM-007 seat-allocation race condition (Section 26 Risks); and duplicate-transaction prevention for payment processing (BR-FEE-006).

19.3 Backup Architecture Detail

Reused directly from Appendix-F's Backup NFR series (NFR-BKP-001 through 005).

NFR ID	Control	Target
NFR-BKP-001	Backup Frequency	Daily full backup + continuous transaction log backup
NFR-BKP-002	Backup Retention Period	30 days rolling + monthly archive for 1 year (Client/Product Decision Required to confirm)
NFR-BKP-003	Recovery Time Objective (RTO)	≤4 hours (Client/Product Decision Required to confirm)
NFR-BKP-004	Recovery Point Objective (RPO)	≤1 hour (Client/Product Decision Required to confirm)
NFR-BKP-005	Backup Integrity Verification	Monthly automated restore test

19.4 Disaster Recovery Scope

A documented, drill-tested failover procedure (NFR-DR-001) is architecturally required for the Database Tier and File/Object Storage Tier (Section 17) at minimum; failover for the Web/Application Tier is comparatively low-risk given its stateless design (Section 17.3), since new instances can be provisioned without data-loss concern. The annual drill cadence (NFR-DR-002) should exercise the full path — database failover, application reconnection, and integration-gateway reconnection — not merely a database-only test.

20. Performance Considerations

Reuses Appendix-F's Performance NFR series (NFR-PERF-001 through 012). Selected targets summarized; full detail in Appendix-F.

Operation	Target
Page Load Time	≤3 seconds (95th percentile)
Login Response	≤2 seconds
Attendance Save	≤1 second per action
Fee Payment Processing	≤5 seconds (gateway confirmation to receipt)
API Response Time	≤500 ms (95th percentile, internal APIs)
Database Query Response	≤200 ms (95th percentile, indexed queries)

20.1 Additional Performance Targets

Operation	Target
Dashboard Loading	≤3 seconds (all role-based widgets)
Search Response	≤2 seconds (student/staff/catalog search)
Report Generation	≤10 seconds (standard reports); background job beyond that (NFR-RPTP-001)
Bulk Import	≤2 minutes per 1,000 records
Export	≤15 seconds up to 10,000 rows
Notification Delivery	≤60 seconds (trigger event to gateway dispatch)

20.2 Performance Design Response

- Indexing strategy (Appendix-H v1.1 Sections 12, 12.3) directly supports the Database Query Response and Search Response targets.
- Caching (NFR-CACHE-001) directly supports the Dashboard Loading and Page Load Time targets for slow-changing master data.
- Background-job processing for large reports and exports (NFR-RPTP-001) ensures the Report Generation and Export targets do not degrade the experience of other concurrent users.

21. Multi-Tenancy / Institute Isolation Strategy

Not Defined in Approved Scope This is carried forward unchanged from Appendix-H v1.1 Section 2.11. RGD v2.0 Section 26 (Pending Decisions) has not confirmed whether the platform must support genuinely separate institutes versus multiple campuses of one institute. The current architecture assumes single-institute; if multi-campus/multi-institute support is confirmed, an `institute_id/campus_id` column pattern would extend the existing schema (Appendix-H v1.1 Section 18.3) without requiring a fundamental redesign, since the polymorphic association pattern already used throughout the schema supports this style of extension.

21.1 If Multi-Institute Support Is Later Confirmed

- Every session-scoped entity (Section 2.9, Appendix-H v1.1) would gain an `institute_id` alongside its existing `academic_session_id`, since a session is currently modeled as global but would need to become institute-specific.
- Role and Permission (Appendix-H ENT-SYS-002) would need to clarify whether roles are shared across institutes or defined per-institute — currently Client/Product Decision Required.
- Reports & Analytics (Section 14) would need an institute-scoping filter added to every dashboard and export, consistent with the same role-scoping pattern already used for Finance/Academic/Management views.
- This is explicitly a Future Expansion item, not a current design gap — the architecture is not broken by the absence of this decision, but it is not free to retrofit either, which is why it is flagged here rather than silently assumed away.

21.2 Single-Institute Working Assumption

For the remainder of this HLD and for Physical/Low Level Design, a single-institute deployment is the working assumption unless and until the client confirms otherwise. This assumption is consistent with, and does not conflict with, RGD v2.0 OBJ-05 ("support multi-campus operation on a single platform") — multi-campus is understood here as multiple physical locations of one institute, sharing one set of Roles/Configuration, which the current architecture already supports without modification; true multi-institute (separate, unrelated schools sharing the platform) is the unconfirmed case.

22. Environment Overview (Development, Testing, UAT, Production)

Environment	Purpose	Data
Development	Active feature development and unit testing	Synthetic/anonymized test data only
Testing (QA)	Integration and system testing against Appendix-E acceptance criteria	Synthetic test data reflecting Appendix-F Scalability targets for load testing
UAT	Client validation against RGD v2.0 and Appendix-E business scenarios	Anonymized production-like data or client-provided sample data (Client/Product Decision Required)
Production	Live operation	Real student/staff/financial data, subject to full Security Architecture (Section 16)

Specific infrastructure sizing, environment promotion process, and CI/CD tooling are Client/Product Decision Required and are Physical Design / DevOps-phase decisions, not fixed in this HLD.

22.1 Environment Promotion Principle

Code and configuration changes are expected to flow Development → Testing → UAT → Production in that order, with each environment progressively more production-like in scale and data sensitivity. No environment other than Production holds real student/staff/financial data, directly supporting the Privacy-by-design posture (NFR-PRIV-001, Appendix-F) and reducing the blast radius of any lower-environment security incident.

22.2 Environment-Specific Considerations

- Development: individual developers or small teams; External Integrations (Section 11) are typically stubbed/mockd rather than connected to live vendor sandboxes.
- Testing (QA): connected to vendor sandbox/test endpoints for External Integrations where available, to validate integration behavior without real transactions.
- UAT: the client's first hands-on exposure to the system against real business scenarios (Appendix-E acceptance criteria); this environment's sign-off is the trigger for the RGD v2.0 Approval Matrix's progression toward go-live.
- Production: subject to the full Security Architecture (Section 16), Availability targets (Section 19), and Backup/DR posture (Section 17) — the only environment where these NFRs are contractually meaningful.

22.3 Environment Parity Considerations

Aspect	Development/Testing	UAT	Production
External Integrations	Mocked/stubbed	Vendor sandbox where available	Live vendor connections
Data Volume	Small synthetic set	Load-test-scale synthetic or anonymized data	Real, growing per Section 18
Security Controls (Section 16)	Relaxed for developer velocity	Full controls enabled for realistic validation	Full controls, mandatory
Availability Target (Section 19)	Best-effort, no SLA	Best-effort, no SLA	99.5% (NFR-AVAIL-001)

23. Traceability to Previous Appendices

Every section of this HLD is derived from, not independent of, the appendices listed in Section 3. No new business requirement, entity, or rule was introduced in producing this document.

HLD Section	Primary Source
5. Major Modules	RGD v2.0 Section 12; Appendix-E
6. Module Interaction	Appendix-D (Module-Wise Dependency Analysis); Appendix-H v1.1 Section 8
8. User Roles	Appendix-E Field 38 (Role Permissions); Appendix-H ENT-SYS-002
9. Authentication & Authorization	Appendix-F NFR-AUTHN/AUTHZ/SESS/PWD series; Appendix-H ENT-SYS-001/002
10. Database Architecture	Appendix-G; Appendix-H v1.1 (full detail, not redesigned)
11. External Integrations	RGD v2.0 Section 15
13. Notification Architecture	Appendix-A Module 9; Appendix-H ENT-COM-002; Appendix-D Dependency Risk Register
15. Logging & Audit	Appendix-F NFR-LOG/AUDIT series; Appendix-H ENT-SYS-004
16. Security Architecture	Appendix-F Security/Authentication/Authorization/Encryption/Network Security NFR series
18-20. Scalability/Availability/Performance	Appendix-F NFR-SCAL/AVAIL/REL/PERF series; Appendix-H v1.1 Section 17
21. Institute Isolation	Appendix-H v1.1 Section 2.11 (reused verbatim in substance)
1-4. Introduction/Scope/References/Overall Architecture	RGD v2.0 Sections 1-2; Appendix-A actor lists
7. Layered Architecture	Appendix-E (functional layer boundary); Appendix-H v1.1 Section 2 (data layer principles)
12. File Storage Architecture	Appendix-G/H ENT-SYS-006 (Document entity)
14. Reporting Architecture	Appendix-A Module 10; Appendix-F NFR-RPTP-001
17. Deployment Overview	RGD v2.0 Section 22 (cloud-hosting assumption); Appendix-F Backup/Availability NFR series
22. Environment Overview	RGD v2.0 Section 22 Business Assumptions; Appendix-F Section 19 test-environment implications
24-26. Assumptions/Constraints/Risks	RGD v2.0 Sections 22-23; Appendix-C v1.1 Section 3; Appendix-D Dependency Risk Register; Appendix-H v1.1 Section 19

23.1 Traceability Completeness Statement

All 27 sections of this HLD trace to at least one of the eight approved source appendices. No section introduces a business fact, entity, rule, or requirement that does not already exist in Appendices A through H v1.1.

24. Assumptions

- Cloud-hosted deployment, per RGD v2.0 Business Assumptions (Section 22) — specific provider Client/Product Decision Required.
- A relational database management system, per Appendix-G/H stated assumption.
- Single-institute operation at go-live, per Section 21 above.
- All external integration vendors (Section 11) remain unselected and are treated as pluggable, replaceable components rather than architecturally fixed dependencies.
- The modular structure already established in Appendix-E/G/H is sufficient to support either a monolithic or services-based physical implementation without HLD-level redesign.
- Load balancing and horizontal scaling of the Web/Application Tier (Section 17) is achievable without architectural change, since no module maintains server-local session state beyond the authenticated User token.
- The client will provide final numeric values for all items currently marked Client/Product Decision Required (concurrency, retention periods, statutory slabs, SLA windows) before Physical Design begins, consistent with the resolution path already recommended in Appendix-C v1.1 Section 4 and Appendix-F's Final Enterprise Review.
- Existing legacy data (if any) will be cleansed by the client before migration, per RGD v2.0 Section 22 Business Assumptions — this HLD does not include a data-migration architecture beyond noting that Bulk Import (FR-07) is the designed ingestion path.

25. Constraints

- This document is technology-neutral by governing instruction — it does not select specific products, frameworks, or vendors.
- No SQL, API specifications, sequence diagrams, class diagrams, source code, deployment scripts, or UI mockups are included, per explicit scope exclusion (Section 2.2).
- This HLD reuses Appendix-H v1.1 as its architectural baseline and does not redesign the logical data model.
- Where the approved appendices do not define an answer, this document states "Client/Product Decision Required" rather than inventing one.
- Diagrams are limited to four simple ASCII forms (Context, Layered Architecture, Module Interaction, Deployment) per the governing instruction — no UML, no vendor-tool-generated diagrams.
- This document cannot resolve the seven external-integration vendor selections (Section 11) or the Institute Isolation Strategy (Section 21) — these require client decisions outside the scope of architectural documentation.

26. Risks

Risk	Source	Architectural Implication
FR-37 (Communication) is a cross-module single point of failure	Appendix-D Dependency Risk Register	Mitigated by independent per-integration health monitoring (NFR-MON-002), not architectural redundancy of the module itself
BR-ADM-001/BR-ADM-007 race condition (seat allocation)	Appendix-C v1.1 Section 3.2	Requires atomic-transaction design at Physical/LLD stage (Section 10 notes this is reused, not resolved, here)
Polymorphic foreign keys cannot be enforced by standard DB constraints	Appendix-H v1.1 Section 19	Requires application-layer integrity checks — an explicit Physical Design task, not assumed solved
Institute Isolation Strategy undefined	Appendix-H v1.1 Section 2.11 / RGD v2.0 Pending Decisions	Current architecture assumes single-institute; retrofitting multi-institute support later is possible but not free
External integration vendors unselected	RGD v2.0 Section 26	Integration Tier (Section 17) is designed as pluggable specifically to absorb this uncertainty
Multi-role individuals (e.g., staff who are also parents)	Section 8.2 (this document)	Handled via separate User accounts rather than combined permissions; requires clear onboarding guidance to avoid confusion
High-volume table growth (AttendanceRecord, AuditLog)	Appendix-F NFR-SCAL series; Appendix-H v1.1 Section 17	Partition strategy (Section 17.8, Appendix-H v1.1) must be implemented from go-live, not retrofitted after volume becomes a problem
Shared Communication service as a cross-module dependency	Appendix-D Dependency Risk Register	Independent per-integration monitoring (NFR-MON-002) mitigates but does not eliminate the architectural coupling

26.1 Risk Ownership

Risk ownership follows the same pattern established in Appendix-D's Dependency Risk Register and Appendix-C v1.1's Section 3: architectural/technical risks (race conditions, polymorphic FK integrity) are owned by the future Solution/Database Architect at Physical Design stage; business-policy risks (Institute Isolation, vendor selection) are owned by the Project Sponsor per RGD v2.0's Approval Matrix.

26.2 Single Point Dependencies (Reused from Appendix-D)

These four requirements/entities were identified in Appendix-D's Critical Dependency List as having very high fan-out with no redundant alternative — their architectural treatment (Section 4.3 Single Source of Truth principle) is the primary mitigation, since true redundancy would mean maintaining duplicate, potentially inconsistent copies of core data.

Element	Why It's Critical	Architectural Mitigation
Student (FR-06 / ENT-SIS-001)	Depended on, directly or indirectly, by effectively all 47 requirements	Single Source of Truth principle (Section 4.3); strict schema change control recommended
Employee (FR-33 / ENT-HR-)	Sole foundation for all HR & Payroll and	Same pattern as Student, at HR & Payroll

Element	Why It's Critical	Architectural Mitigation
001)	staff-side Attendance	module scope
Admission Application (FR-01)	Sole entry point into the entire student lifecycle	No architectural redundancy provided; a Business Continuity workaround exists instead (NFR-BC-001)
Communication Service (FR-37)	Shared notification service for 4+ modules; failure silently disables all of them at once	Independent per-integration monitoring (NFR-MON-002), not structural redundancy

27. Validation Summary

Check	Result
All 27 required sections present	PASS — Sections 1 through 27 as specified
No Low Level Design content included	PASS — no algorithms, pseudocode, or module-internal design
No SQL/DDL included	PASS
No API specifications included	PASS
No sequence or class diagrams included	PASS — only Context, Layered Architecture, Module Interaction, and Deployment diagrams, all in simple ASCII form
No new business requirements invented	PASS — every module, role, entity, and NFR reused by direct reference to Appendices A-H v1.1
Technology-neutral	PASS — no specific vendor, framework, or cloud provider is named as a fixed decision; all are flagged Client/Product Decision Required where relevant
Reuses Appendix-H v1.1 as architectural baseline without redesign	PASS — Section 10 explicitly summarizes rather than re-specifies
All 71 Business Rules (Appendix-C v1.1) architecturally addressed	PASS — enforcement mechanism defined in Section 7.1; summary in Section 16.3
All 40 NFR categories (Appendix-F) architecturally addressed	PASS — full mapping table in Section 4.5
All 11 modules (RGD v2.0 / Appendix-E) represented	PASS — Section 5, with individual descriptions in Section 5.1.1 through 5.1.11
All external integrations (RGD v2.0 Section 15) represented	PASS — Section 11, all 7 integrations listed with status
Every Client/Product Decision Required item is a genuine open decision, not a disguised gap	PASS — cross-checked against the same open items already tracked in Appendix-C v1.1 Section 3.5, Appendix-F, Appendix-G, and Appendix-H v1.1

Overall HLD Quality Score

90/100 — deductions reflect the volume of infrastructure and vendor decisions still pending client confirmation (hosting provider, integration vendors, institute isolation strategy), consistent with the same open items already tracked across Appendix-C v1.1, Appendix-F, and Appendix-H v1.1. No deduction is taken for structural completeness, which is full.

Approval Status

Conditionally Ready — full architectural coverage across all 27 required sections with zero fabricated requirements and full traceability to Appendices A-H v1.1; approval is gated on the client resolving the consolidated Client/Product

Decision Required items identified throughout (primarily: hosting/infrastructure provider, external integration vendor selections, and Institute Isolation Strategy confirmation).